

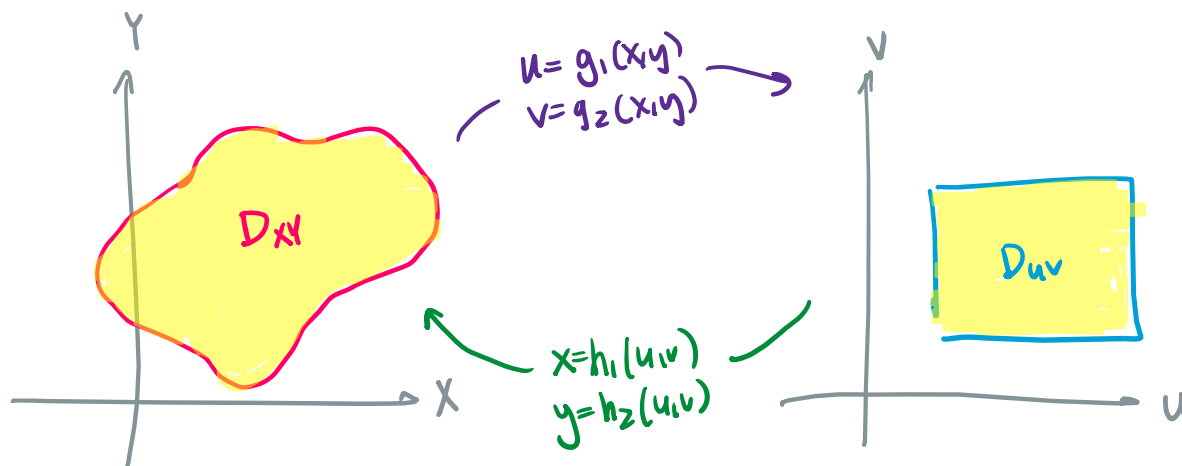
6.7: Joint Distributions of Functions of Random Variables

Let X and Y be jointly continuous random variables with density function f_{XY} and define $U = g_1(X, Y)$, $V = g_2(X, Y)$ to be the random variables that are functions of X and Y .

If both of the following conditions are met:

- (i) $u = g_1(x, y)$ and $v = g_2(x, y)$ can be solved by $x = h_1(u, v)$ and $y = h_2(u, v)$
- (ii) $J_{XY \rightarrow UV} := \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \neq 0$ (and all such derivatives exist)

Then $f_{UV}(u, v) = f_{XY}(h_1(u, v), h_2(u, v)) \cdot |J_{XY \rightarrow UV}|$



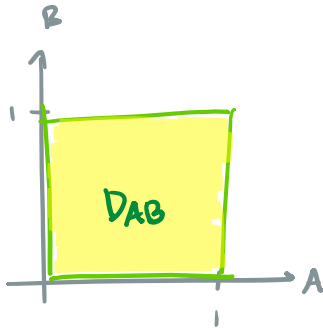
$$J_{XY \rightarrow UV} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u$$

$$J_{UV \rightarrow XY} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = u_x v_y - u_y v_x$$

$$\iint_{D_{XY}} f(x, y) dA = \iint_{D_{UV}} f(x(u, v), y(u, v)) \cdot |J_{XY \rightarrow UV}| dA$$

$$\iint_{D_{UV}} f(u, v) dA = \iint_{D_{XY}} f(u(x, y), v(x, y)) \cdot |J_{UV \rightarrow XY}| dA$$

Example: Let A and B be uniform random variables on the interval $(0, 1)$, and define $U = 2A + B, V = 5A + B$. Find $f_{U,V}$



$$f_{AB}(a,b) = \begin{cases} 1 & 0 \leq a \leq 1 \\ & 0 \leq b \leq 1 \\ 0 & \text{else} \end{cases}$$

$$\begin{aligned} (U &= 2A + B) \\ -(V &= 5A + B) \\ \hline U - V &= -3A \end{aligned}$$

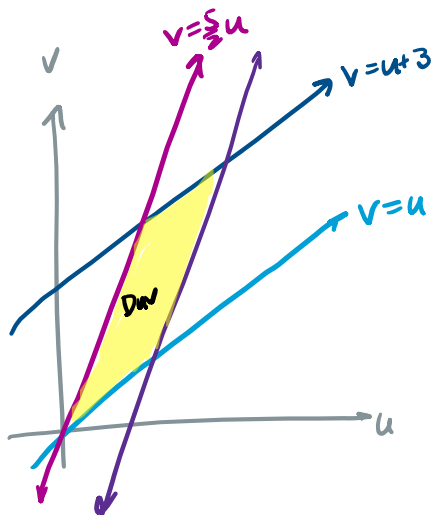
$$A = \underline{\underline{-\frac{1}{3}U + \frac{1}{3}V}}$$

$$\begin{aligned} 5(U &= 2A + B) \\ -2(V &= 5A + B) \\ \hline 5U - 2V &= 3B \end{aligned}$$

$$B = \underline{\underline{\frac{5}{3}U - \frac{2}{3}V}}$$

$$J_{AB \rightarrow UV} = \begin{vmatrix} A_u & A_v \\ B_u & B_v \end{vmatrix} = \begin{vmatrix} -1/3 & 1/3 \\ 5/3 & -2/3 \end{vmatrix} = (-1/3)(-2/3) - 1/3 \cdot 5/3 = \frac{2}{9} - \frac{5}{9} = -\frac{1}{3}$$

$$f_{UV}(u,v) = f_{AB}(a(u,v), b(u,v)) \cdot |J_{AB \rightarrow UV}| = \begin{cases} 1 \cdot \left| -\frac{1}{3} \right| & 0 \leq -\frac{1}{3}u + \frac{1}{3}v \leq 1 \\ & 0 \leq \frac{5}{3}u - \frac{2}{3}v \leq 1 \\ 0 & \text{else} \end{cases}$$

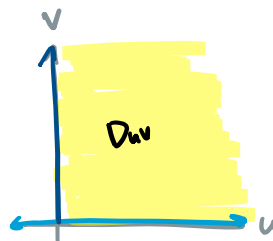
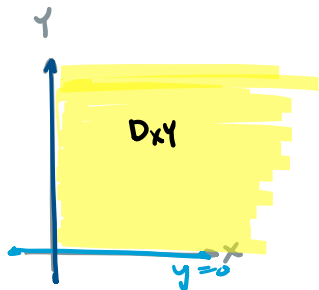


$$\begin{aligned} 0 &= -\frac{1}{3}u + \frac{1}{3}v \leftarrow v = u \\ 1 &= -\frac{1}{3}u + \frac{1}{3}v \leftarrow v = u + 3 \\ 0 &= \frac{5}{3}u - \frac{2}{3}v \leftarrow v = \frac{5}{2}u \\ 1 &= \frac{5}{3}u - \frac{2}{3}v \leftarrow v = \frac{5}{2}u - \frac{3}{2} \end{aligned}$$

$$= \begin{cases} \frac{1}{3} & \text{in } D_{UV} \\ 0 & \text{else} \end{cases}$$

Example: Let X and Y have joint density $f_{XY}(x, y) = \frac{1}{6}xye^{-y-\frac{x}{y}}$ for $x, y > 0$.

If we define $U = \frac{X}{Y}$, $V = Y$, evaluate $E[U^n] = E\left[\left(\frac{X}{Y}\right)^n\right]$



$$\begin{aligned} y=0 &\rightarrow v=0 \\ x=0 &\rightarrow uv=0 \\ &\downarrow \\ &u=0 \text{ when } v>0 \end{aligned}$$

$$u = \frac{x}{y} \rightarrow u = \frac{x}{y} \rightarrow x = uv$$

$$v = y \rightarrow y = v$$

$$J_{XY \rightarrow UV} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

$$= \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v \cdot 1 - u \cdot 0 = v$$

$$f_{UV}(u, v) = f_{XY}(x(u, v), y(u, v)) / |J_{XY \rightarrow UV}|$$

$$= \frac{1}{6} \cdot uv \cdot v \cdot e^{-v-\frac{uv}{v}} \cdot |v|$$

$$= \frac{1}{6} uv^3 e^{-v-u} \quad \left(\begin{matrix} u > 0 \\ v > 0 \end{matrix} \right)$$

$$E[U^n] = \int_{D_{UV}} u^n \cdot f_{UV}(u, v) dA$$

$$= \int_0^\infty \int_0^\infty u^n \cdot \frac{1}{6} uv^3 e^{-v-u} du dv$$

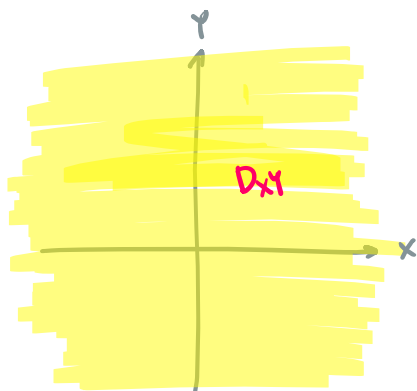
$$= \frac{1}{6} \int_0^\infty u^{n+1} e^{-u} du \cdot \int_0^\infty v^3 e^{-v} dv$$

$$= \frac{1}{6} \int_0^\infty u^{(n+2)-1} e^{-u} du \cdot \int_0^\infty v^{4-1} e^{-v} dv$$

$$= \frac{1}{6} \cdot \Gamma(n+2) \cdot \Gamma(4) = \frac{1}{6} \cdot (n+1)! \cdot 3!$$

$$\boxed{= (n+1)!}$$

Example: Let X, Y be independent standard normal random variables and denote (R, Θ) to be the equivalent polar form of the rectangular point (X, Y) . Find $f_{R, \Theta}$ and $E[R] = E[\sqrt{X^2 + Y^2}]$

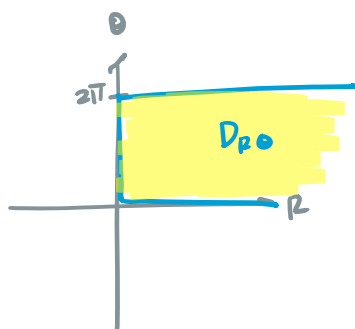


$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}$$

$$f_{XY}(x, y) = f_X \cdot f_Y = \frac{1}{2\pi} e^{-\frac{x^2}{2} - \frac{y^2}{2}}$$

$$\begin{aligned} x &= R \cos \theta \\ y &= R \sin \theta \end{aligned}$$

$$\begin{aligned} J_{XY \rightarrow R\Theta} &= \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} \\ &= r \cos \theta \cdot \cos \theta - \sin \theta \cdot (-r \sin \theta) \\ &= r \end{aligned}$$



$$\begin{aligned} f_{R, \Theta}(r, \theta) &= f_{XY}(x(r, \theta), y(r, \theta)) \cdot |J_{XY \rightarrow R\Theta}| \\ &= \frac{1}{2\pi} e^{-\frac{(r \cos \theta)^2}{2} - \frac{(r \sin \theta)^2}{2}} \cdot |r| \end{aligned}$$

$$= \frac{1}{2\pi} r e^{-r^2/2} \quad (\text{Fr } 0 \leq r < \infty, 0 \leq \theta < 2\pi)$$

$$E[R] = \iint_{D_{R\Theta}} r \cdot f_{R\Theta}(r, \theta) dA = \int_0^{2\pi} \int_0^\infty r \cdot \frac{1}{2\pi} r e^{-r^2/2} dr d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} d\theta \cdot \int_0^\infty r^2 e^{-r^2/2} dr = \frac{\sqrt{2\pi}}{2} \int_{-\infty}^\infty r^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{r^2}{2}} dr$$

$$= \frac{\sqrt{2\pi}}{2} \cdot E[Z^2] \quad \text{Fr } Z \sim N(0, 1)$$

$$= \frac{\sqrt{2\pi}}{2} \cdot [\underbrace{\text{Var}(Z)}_1 + \underbrace{(E[Z])^2}_{0^2}] = \boxed{\sqrt{\frac{\pi}{2}}}$$

Example: Let $X_1, X_2, \dots, X_n$ be independent copies of $\text{Exp}(\lambda)$ and define $Y_m := \sum_{k=1}^m X_k$

Find the joint density function of $Y_1, Y_2, \dots, Y_n$ and then the density f_{Y_n}

$$f_{X_k}(x_k) = \lambda e^{-\lambda x_k} \quad (x_k \geq 0) \rightarrow f_{X_1, X_2, \dots, X_n}(x_1, \dots, x_n) = \lambda^n e^{-\lambda(x_1 + x_2 + \dots + x_n)} \quad \begin{pmatrix} x_1 \geq 0 \\ x_2 \geq 0 \\ \vdots \\ x_n \geq 0 \end{pmatrix}$$

$$\begin{aligned} Y_1 &= X_1 \\ Y_2 &= X_1 + X_2 \\ &\vdots \\ Y_n &= X_1 + \dots + X_n \end{aligned}$$

$$J_{Y_1, \dots, Y_n - X_1, \dots, X_n} = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 1 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1 \end{vmatrix} = 1$$

$$\begin{aligned} X_1 &= Y_1 \\ X_2 &= Y_2 - Y_1 \\ &\vdots \\ X_n &= Y_n - Y_{n-1} \end{aligned}$$

$$J_{X_1, \dots, X_n - Y_1, \dots, Y_n} = \frac{1}{1} = 1$$

$$\begin{aligned} f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) &= f_{X_1, \dots, X_n}(y_1, y_2 - y_1, \dots, y_n - y_{n-1}) \cdot |1| \\ &= \lambda^n e^{-\lambda(y_1 + (y_2 - y_1) + \dots + (y_n - y_{n-1}))} = \lambda^n e^{-\lambda y_n} \quad (0 \leq y_1 \leq y_2 \leq \dots \leq y_n) \end{aligned}$$

$$f_{Y_n}(y) = \int \int \dots \int_{0 \leq y_1 \leq y_2 \leq \dots \leq y} f_{Y_1, \dots, Y_n}(y_1, y_2, \dots, y) dA$$

$$= \int_0^y \int_0^{y_{n-1}} \dots \int_0^{y_3} \int_0^{y_2} \lambda^n e^{-\lambda y} dy_1 dy_2 \dots dy_{n-1}$$

$$= \lambda^n e^{-\lambda y} \int_0^y \int_0^{y_{n-1}} \dots \int_0^{y_3} y_1 dy_2 \dots dy_{n-1}$$

$$\underbrace{\frac{1}{2} y_2^2 \Big|_0^{y_3} = \frac{1}{2} y_3^2}_{\underbrace{\frac{1}{6} y_3^3 \Big|_0^{y_4} = \frac{1}{6} y_4^3}}$$

$$= \lambda^n e^{-\lambda y} \cdot \frac{1}{(n-1)!} y^{n-1}$$

$$= \boxed{\frac{\lambda^n}{(n-1)!} y^{n-1} e^{-\lambda y} \quad (y \geq 0)} \leftarrow \text{GAMMA}$$